

1. Administrative & Study Identification

Study Title: A Study Of Palbociclib (PD-0332991) + Letrozole VS. Placebo+ Letrozole For 1st Line Treatment Of Asian Postmenopausal Women With ER+/HER2- Advanced Breast Cancer [PALOMA-4] **Official Full Study Title:** A MULTICENTER, RANDOMIZED, DOUBLE-BLIND PHASE 3 STUDY OF PALBOCICLIB (ORAL CDK 4/6 INHIBITOR) PLUS LETROZOLE VERSUS PLACEBO PLUS LETROZOLE FOR THE TREATMENT OF PREVIOUSLY UNTREATED ASIAN POSTMENOPAUSAL WOMEN WITH ER (+), HER2 (-) ADVANCED BREAST CANCER **Short Title/Acronym:** PALOMA-4 **Protocol Number:** A5481027 **NCT Number:** NCT02297438 **Posting ID:** 315791112436234872 **Trial Status:** COMPLETED **Sponsor/Company Name:** Pfizer **Investigational Product:** Palbociclib (PD-0332991) + Letrozole **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the clinical benefit, specifically Progression-Free Survival (PFS), following treatment with letrozole in combination with palbociclib versus letrozole in combination with placebo in Asian postmenopausal women with ER(+)/HER2(-) advanced breast cancer who have not received prior systemic anti-cancer therapies for their advanced/metastatic disease. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) based on Blinded Independent Central Review (BICR), Objective Response (OR), overall survival, and the safety and tolerability profile.

2.2 Background & Rationale To confirm the efficacy and safety of the cyclin-dependent kinase 4/6 (CDK4/6) inhibitor palbociclib combined with the aromatase inhibitor letrozole in the specific demographic of Asian postmenopausal women with ER+/HER2- Advanced Breast Cancer (ABC). While the combination has demonstrated a manageable safety profile and strong efficacy in international phase 3 trials (e.g., PALOMA-2), this study aims to validate those outcomes and provide robust real-world comparable evidence for this targeted ethnic subpopulation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1 ratio)

3.2 Planned Enrollment & Duration Target \$\$\$: 340 participants
Number of Sites: Multicenter (Across China and other Asian regions)
Estimated Study Start: 23-Mar-2015
Estimated Primary Completion: 31-Aug-2020

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult Asian women with locoregionally recurrent or metastatic disease not amenable to curative therapy. 2. Confirmed diagnosis of Estrogen Receptor-positive (ER+) breast cancer. 3. Postmenopausal women with no prior systemic anti-cancer therapy for advanced ER+ disease. 4. Measurable disease as per RECIST v1.1 or bone-only disease, with an Eastern Cooperative Oncology Group [ECOG] performance status of 0-1. 5. Adequate organ and marrow function. 6. Patient must agree to provide tumor tissue.

4.2 Exclusion Criteria 1. Confirmed diagnosis of HER2-positive disease. 2. Patients with advanced, symptomatic visceral spread at risk of life-threatening complications in the short term, or known uncontrolled/symptomatic CNS metastases. 3. Prior treatment with any CDK 4/6 inhibitor, or prior neoadjuvant/adjuvant treatment with a non-steroidal aromatase inhibitor (e.g., anastrozole or letrozole) with disease recurrence while on or within 12 months of completing treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Palbociclib 125 mg taken once daily for 21 consecutive days followed by 7 days off (28-day cycle), in combination with Letrozole (Trade Name: Femara; ATC Code: L02BG04) 2.5 mg taken once daily continuously throughout the 28-day cycle. The comparator arm receives Placebo + Letrozole 2.5 mg. **Route of Administration:** Oral **Duration of Treatment:** Continued until objective disease progression, unacceptable toxicity, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for comorbidities and bone-modifying agents. **Prohibited:** Concurrent systemic anti-cancer therapies, prior CDK 4/6 inhibitors, and strong CYP3A inhibitors/inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Assessment (RECIST v1.1), Safety Labs
Baseline	Day 0	Randomization, First Dose of Investigational Product, Dispensing
Interim	Every 4 Weeks	Safety Labs (CBCs for neutropenia), AE Assessment, IP

Accountability. Tumor imaging every 8-12 weeks | | **End of Study** |
At Progression | Final Tumor Scan, Exit Interview, Safety Follow-
Up, Transition to Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) Based on Investigator's Assessment. Defined as the time from the date of randomization to the date of the first documentation of objective progression of disease (PD) or death due to any cause in the absence of documented PD. **Measurement Method:** Investigator's assessment of radiological imaging using Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 as a 20% increase in the sum of diameters of target measurable lesions above the smallest sum observed, with a minimum absolute increase of 5 mm.

7.2 Secondary & Exploratory Endpoints 1. Progression-Free Survival (PFS) Based on Blinded Independent Central Review (BICR). 2. Percentage of Participants With Objective Response (OR) Based on Investigator Assessment. 3. Percentage of Participants With Objective Response (OR) Based on Investigator Assessment (Participants With Measureable Disease at Baseline). 4. Disease Control/Clinical Benefit Response (DC/CBR) defined as CR, PR, or stable disease ≥ 24 weeks. 5. Safety and tolerability profile (e.g., evaluating frequency and severity of adverse events like neutropenia and leukopenia).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical visits and hematology labs. Special protocol focus on myelosuppression (Grade 3/4 neutropenia, leukopenia, anemia) which is a known class effect of CDK4/6 inhibitors. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) chartered to periodically review safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy endpoints. Safety Set for all participants who received at least one dose of the study drug. **Sample Size Justification:** Target $N=340$ powered to detect a statistically significant improvement in median PFS (Hazard Ratio) between the treatment and placebo arms. **Handling of Missing Data:** Time-to-event endpoints (e.g., PFS) analyzed using Kaplan-Meier estimates. Patients without a documented event at

the time of data cutoff are right-censored at the date of their last valid tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol compliance and patient safety. **Audit Trail:**

Mandatory eCRF locking, source data verification (SDV), and query resolution via standard data clarification forms.

11. Study Identifiers & Governance

Primary ID: A5481027 **Registry Number:** NCT02297438 **Sponsor/Lead Organization:** Pfizer **Posting ID:** 315791112436234872 **Trial Status:** COMPLETED **Study Title:** PALOMA-4 **Official Regulatory Title:** A MULTICENTER, RANDOMIZED, DOUBLE-BLIND PHASE 3 STUDY OF PALBOCICLIB (ORAL CDK 4/6 INHIBITOR) PLUS LETROZOLE VERSUS PLACEBO PLUS LETROZOLE FOR THE TREATMENT OF PREVIOUSLY UNTREATED ASIAN POSTMENOPAUSAL WOMEN WITH ER (+), HER2 (-) ADVANCED BREAST CANCER

12. Clinical Framework (Breast Cancer Specific)

Primary Condition / Disease Area: Breast Neoplasms (Advanced ER-positive, HER2-negative breast cancer) **Diagnosis Threshold:**

Histologically confirmed ER-positive and HER2-negative locoregionally recurrent or metastatic breast cancer **Medical**

Methodology: Targeted CDK4/6 Inhibitor combined with Non-steroidal Aromatase Inhibitor (Endocrine Therapy) **Optimization**

Target: Delaying disease progression (Maximizing PFS) and tumor shrinkage

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced/metastatic breast cancer **Education Protocol (HCP):** Site Initiation Visits (SIV), GCP, and RECIST v1.1 standardized training

Education Protocol (Patient): Oral drug administration instructions, dosing diary maintenance, and infection/AE signs self-monitoring **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of tumor scans and routine IP

accountability tracking

14. Observation & Follow-Up Schedule

Total Duration: Followed until disease progression, death, or study termination **Onsite Visit Cadence:** Every 4 weeks (Day 1 of each 28-day cycle) **Remote Monitoring Frequency:** Survival follow-up conducted every 12 weeks post-disease progression **Checklist Review Interval:** Radiological tumor assessments every 8-12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				